Lab 5: AWK

In this exercise, use the `capitals.txt` provided with this lab to process it using `awk` and shell scripts. The goal is to learn how to create quick `awk`-scripts that process data from a file and format it or do calculations on the fly. Refer to the `awk` manual page to get help on parameters and general use.

Solve the following exercises in order given here. Build on your previous scripts and enhance them further, but make sure to keep a copy around in case you have to start over.

1. Check the top lines of the file using `head`. The first line is a comment describing the content of each column. Write an `awk` script that outputs a descriptive text (column headers) at the beginning and then prints the file contents itself. After printing the contents, let `awk` print a separate line stating `end of file`.
2. The next iteration of your script should output the column headers again after printing 30 lines. Write a function for printing the column headers and call it each time after reaching 30 lines.
3. Use `awk`'s built-in variables to print a summary after the `end of file`. Output filename and total number of lines in the file.
4. In `awk`'s man page, you will find how to specify a custom field separator instead of the default one (space). Use it to remove the separators (`,`) between the columns and print the file contents in tabular format.
5. Concentrate more on the actual data within the file now. Use `awk` combined with a shell script (when needed) to answer the following questions:
    - What is the total number of citizens in all capitals?
    - What is the name of the city with the biggest/smallest population?
    - Which continent has the most capitals?
    - How many citizens live in the capitals for each continent?

Working with dates

Friday the 13th is considered an unlucky day by superstitious people. Regardless of whether you believe in it or not, you are to determine the following: output all years and the corresponding months for the next 50 years in which the 13th of a month falls on a Friday. Output these data in tabular form. You can obtain a calendar using the program `cal` or `ncal`.

Use a combination of `grep` and `awk` to find these special dates.

Blood Donation Calculator

Susan Sunshine is looking forward to her 18th birthday on May 1, 2026, because then she is allowed to legally donate blood for the first time in Germany. She knows the following facts about blood donation:

- Healthy people can donate blood up to six times per year.
- A rest period of two months must be observed between two donations, although three are recommended. The minimum interval between two whole-blood donations in Germany is 56 days (eight weeks).

Write a script that answers the following questions:

- How many blood donations can Susan complete in total, assuming that she is always physically healthy and has time to do so, up to her 70th birthday? Calculate the respective dates of the blood donation appointments by taking into account the facts mentioned above. Output the calculated values in the following format:

```
Donation no. Date       Age
1            01.05.2026 18
2            ...        ...
```

With the command `date`, Unix timestamps (seconds since 1 Jan 1970) can be generated using the parameter `-r`, which you can use to calculate new dates by addition and output them in a formatted way.

- Extend the blood donation program so that Susan does not donate blood on a Friday the 13th (using your program above) and instead donates on the following day. Note this in the output with an extra line in the form:

```
Donation on: 13.11.2026 falls on a Friday, next donation: 14.11.2026
```

- Assuming Susan receives an expense allowance of 8 euros for each donation. For each anniversary donation (10th, 20th, 25th, 50th, 75th and 100th donation) she instead receives exactly that amount (e.g. 25 euros for the 25th donation). How much money has Susan received in total? Extend the output of the program for the money amount received and a total:

```
Donation No. Date       Age   Amount received Total
1            01.05.2026 18            8 Euros 8 Euros
2            ...        ...           8 Euros 16 Euros
3            ...        ...           ...     ...
```

- Extend the program so that the start date of the first donation and the maximum age can be passed to the shell script as parameters on the command line.